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Security Development Lifecycle (SDL) for IC…



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ICS security best practice guidelines for ENISA and the ISA 99. He also frequently lectures to private and public audiences around the world.

The Security Development Lifecycle (SDL) is a software development process that helps developers build more secure software and address security compliance requirements while reducing development cost. The cost part is important to consider as the later you introduce security in the process the more expensive it turns to fix issues, and it gets more difficult and time consuming.

The SDL involves modifying a software development organization's processes by integrating measures that lead to improved software security. The intention of these modifications is not to totally overhaul the process, but rather to add well-defined security checkpoints and security deliverables.

The main principles for building secure software on which SDL is based are Secure by Design, Secure by Default, Secure in Deployment and Secure communications (SD3 + C):

Secure by Design

The software should be architected, designed, and implemented so as to protect itself and the information it processes, and to resist attacks.

Secure by Default

In the real world, software will not achieve perfect security, so designers should assume that security flaws would be present. To minimize the harm that occurs when attackers target these remaining flaws, software's

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activities associated with the standard phases of the Software development Lifecycle are defined, the main points being:

Requirements:

Establish security requirements of the product/service
Create quality gates or bug bars
Security & privacy risk assessment of the product/service

Design:

Establish design requirements
Analysis of the attack surface
Threat Modeling

Implementation:

Use approved tools
Deprecate unsafe functions
Static analysis

Verification:

Dynamic analysis

Fuzz testing
Attack surface review

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It should be clear from the above that the consequences of insecure software can put peoples lives in danger, potentially causing millions of dollars in losses due to shutting down the critical infrastructure of a country due to a single software vulnerability.

It is encouraging to see a number of ICS vendors now making the move to on-board SDL into their Software development lifecycle with the objective of providing secure products to increase critical infrastructure security.

We believe that SDL adoption is an activity that will provide the bests results for a company that is looking to produce secure software/services, as we mentioned before the sooner we address security in the development life cycle the cheaper and easier it will be.

We at Applied Risk can help you to implement SDL in your organization, if you are interested in learning more about SDL for ICS/SCADA product please contact us!

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